

Is AI chatbot recommendation convincing customer? An analytical response based on the elaboration likelihood model

Xiaoyi Zhang^a, Angelina Lilac Chen^b, Xinyang Piao^c, Manning Yu^d, Yakang Zhang^e,
Lihao Zhang^{f,*}

^a College of Liberal Arts and Science, University of Illinois Urbana-Champaign, 702 S. Wright St., MC-448, Urbana, 61801, IL, USA

^b Le Regent School, Rue du Zier 4, 3963, Crans-Montana, Switzerland

^c Electrical Engineering Department, Columbia University, 500 W. 120th Street, New York 10027, NY, USA,

^d Department of Statistics, Columbia University, 1255 Amsterdam Avenue, New York 10027, NY, USA

^e Industrial Engineering and Operations Research Department, Columbia University, 500 W. 120th Street, New York 10027, NY, USA

^f Department of Information Engineering, 8th Floor, Ho Sin Hang Engineering Building, The Chinese University of Hong Kong, Shatin, N.T., Hong Kong

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ABSTRACT

The integration of artificial intelligence (AI) technology in e-commerce has currently stimulated scholarly attention, however studies on AI and e-commerce generally relatively few. The current study aims to evaluate how artificial intelligence (AI) chatbots persuade users to consider chatbot recommendations in a web-based buying situation. Employing the theory of elaboration likelihood, the current study presents an analytical framework for identifying factors and internal mechanisms of consumers' readiness to adopt AI chatbot recommendations. The authors evaluated the model employing questionnaire responses from 411 Chinese AI chatbot consumers. The findings of present study indicated that chatbot recommendation reliability and accuracy is positively related to AI technology trust and have negative effect on perceived self-threat. In addition, AI technology trust is positively related to intention to adopt chatbot decision whereas perceived self-threat negatively related to intention to adopt chatbot decision. The perceived dialogue strengthens the significant relationship between AI-tech trust and intention to adopt chatbot decision and weakens the negative relationship between perceived self-threat and intention to adopt AI chatbot decisions.

1. Introduction

Over the last few years, AI has been integrated into business processes and management, workplaces and organizational structures, and social dialogs and communications, as well as redefining the ways people engage (Ali et al., 2024; Wong et al., 2024). Many artificial intelligence (AI) chatbots, i.e., Anna (IKEA), Louis (Air France), Connie (Hilton), and Alixiaomi (Taobao platform) have turned into standard software in customer care (Luo et al., 2019; Nichifor et al., 2021). A market insights report by Marketandmarkets (2024) reported that market values of AI chatbots is expected to exceed \$15.5 billion in 2028, as compared to \$250 million in 2019 (Luo et al., 2019). The rise follows a major reason of time and cost reduction (Adam et al., 2021). Given that the AI chatbots are incorporated regularly in the e-commerce organizations and interface directly with the client during the purchasing process (Pillai et al., 2024), customer service application has gained

popularity. Nonetheless, two main concerns persist, and these are the central areas of concern in this research. Firstly, as an important part of customer service, recommending a good product is essential to the customer. On-line product recommendations have enormous impact both in practice and research in the e-commerce field (Xiao & Benbasat, 2007). In light of AI chatbots taking the lead on the e-commerce front, they now perform personalized product recommendations. In contrast, the AI chatbot recommendations have a specific content and delivery style that creates an obstacle in apprehending the consequences they have on the customer's readiness to adopt them (Xiao & Benbasat, 2007). Secondly, the customer adoption of the AI chatbots is still an untapped potential (Cheng et al., 2022). Studies show that shoppers prefer to communicate with human agents (Shumanov & Johnson, 2021). Nevertheless, it remains unclear which aspects of AI chatbot communication can influence a user's interest in adopting a decision through a chatbot (Pillai et al., 2024). Furthermore, customer AI

* Corresponding author.

E-mail address: lhzhanguh@ieee.org (L. Zhang).

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chatbots' adoption decision remains an intention that needs further research. Earlier research has indicated that customers often expect the human aspect in service provision (M. Li & Wang, 2023; Pillai et al., 2024; Sidlauskienė et al., 2023). It is also noteworthy that the revelation of AI engagement usually lowers customers' desire to purchase the product (Luo et al., 2019). As a result, it is challenging to craft AI chatbot conversations interesting to clients and convince them to adopt chatbot decisions. Optimizing the efficiency of AI chatbot conversation to increase a customer's intention to use chatbots is an issue for organizations that deploy AI chatbots for the communication.

Our journey in the context of AI chatbot recommendations from an online shopping perspective involves previewing behavioral patterns and empirical-conceptual perspectives regarding users. One of the principal studies in this regard include those completed by Sheehan et al. (2020) and Cheng et al. (2022) that explore the bottom of the line of the psychological and behavioral processes in chatbot interactions close to humans. In the course of conducting experiments, the interaction between the humans and the AI chatbots was examined in terms of adoption, continued use intention, purchase intent, purchase behavior, anthropomorphism, trust, affective attachment to AI chatbots, and satisfaction (Sheehan et al., 2020). By exploring the role of customer-level characteristics like one's desire to communicate with a fellow human being and how it affects expectations and emotions, researchers hope to discover the hidden motives through which attitudes, intended behaviors, and actual behaviors concerning AI chatbots are shaped.

With respect to the use of AI chatbot adoption domain a major research gap occurs referring to effect of AI chatbot recommendations on consumer intention to apply these virtual assistants (Pillai & Sivathanu, 2020). AI chatbots are being used more and more across a wide range of customer service contexts, but there is not much reliable scientific evidence, which probed the effect of recommendations coming from these chatbots among consumers on their attitude towards those chatbots and their intention to adopt them. It is important to address the space in the literature, especially that the recommendations have a central role in this decision behavior, especially in a situation where AI chatbots become more and more popular on the internet. Through this line of inquiry, in our research we meant to close this gap by providing an empirical investigation of the effect of AI chatbot suggestions on consumer intent to adopt this technology. Our plan will be to carry out survey-based study, with the focus on examining the AI chatbot instant recommendations and their accuracy and reliability, or personalized guidance, on changing the consumer perception and attitude towards adoption of AI chatbots. Furthermore, the investigation is designed to provide insights on the triggering psychological processes as well as moderators that may either promote or inhibit the recommendation influence on adoption decisions. The main aim of this study is to fill the research gap and give important insights to business and practitioners which will improve the way AI chatbots are carried out in customer service thus helping promote their acceptance and adoption by the consumers.

In our exploration, we make use of the elaboration likelihood model (ELM) as our theoretical framework on factors to determine consumer's intention to use AI chatbots. ELM displays a holistic representation of mental activities engaged in the process of persuasion and decision-making (Petty et al., 1986), that most notably are very important during the digital era. By integrating ELM into our study, we aim to explore how individuals process information about chatbot recommendation reliability and accuracy along two distinct routes: the route through the inner mind or the superficial route which is based on the simple cues and evokes less elaboration. In our proposed model, chatbot reliability and accuracy have been established as the two independent variables which then, have a direct effect on the level of intention to adopt AI chatbots. Given our focus on mediating factors including AI technology trust and perceived self-threat which may determine how individuals engage in elaboration process is paramount. Moreover, the moderating effect of perceived dialogue is expected to influence the strength of relationship

between AI technology trust, and perceived self-threat towards intention to adopt AI chatbot decisions. Our research falls under this conceptual model with the aim of moving beyond the traditional view of the cognitive processes at work, trust, threat perception, dialogue perception, and finally, the decision to adopt AI chatbots. Fig. 1 indicated the suggested model of the study.

2. Theoretical background

2.1. Elaboration likelihood model theory

The Elaboration Likelihood Model (ELM), which was initially conceived by Petty et al. (1981), represents a significant dual-process model that accounts for how attitudes change and evolve. This model, according to J. Kitchen et al. (2014), explains the complicated and sophisticated processes people undergo in their attitudinal changes through communication. By virtue of ELM, it was established that external information influences attitude and subsequent behavior modification. This type of information informing individuals opens their option and makes them to alter a previous stigma, perception, and potentially the current behavior. When individuals encounter persuasive stimuli, they traverse two distinct paths: the central route and the peripheral route. The main route featuring the cognitive processing plays a crucial role whereby the recipients have to exercise in-depth analysis of the message contents which enhances the attitude change process carefully. Alternatively, the peripheral route assumes the use of passive processes, where people pay attention to facial cues, gender of the communicator, status hierarchy and the like. The decisive role in persuasion through central route mostly depends on the more or less high cognitive level of the recipients and their adequate engagement in the message, as asserted by Petty and Cacioppo (1984) and Petty et al. (1986). The dual-process model under the name of Elaboration Likelihood (EL) which constitutes the cognitive capacity and motivation eases the flow of messages into the system. Among the recipients with a high level of EL, they usually undertake refined information sorting tasks, which may lead to an attitude change as suggested by C.-Y. Li (2013) and J. Kitchen et al. (2014). These central path cognitions, rebuttals, act as a mediator for the personal assessment of attitude objectiveness which as the central path in the persuasion model have been identified by Bhat-tacherjee and Sanford (2006). Based on an individual's motivation and abilities, their probability of elaboration will be either high or low, determining the path of convincing. In accordance with the ELM theoretical terms, whenever an AI Chatbot provides suggestion information to a client, the customer activates the cognitive system and uses its cognitive capacity to assess the information integrity of the suggested content, affecting customers via the central channel. Furthermore, AI chatbot exhibits humanistic traits, which are the most crucial aspect of AI-human connection.

In the e-commerce context, central cues depend on the determination of content legitimacy based on the theory set forth by Petty and Cacioppo (1984) as well as the advantage of a product over other brands that is illustrated in the study done by Mousavizadeh et al. (2022) and the outcome produced with product use, as discussed by MacInnis and Stayman (1993). The idea of central cues, intended in the subject of our AI chatbot adoption research, these variables could be chatbot recommendation reliability, accuracy and perceived dialogue, which are people's appraising and forming cognitive process and the intention to adopt after that.

In case of limited cognitive ability belongs to the ELM, individuals may be strongly influenced by peripheral cues instead of deep understanding of the information (Mousavizadeh et al., 2022). According to this analogy, individuals with limited cognitive capacity could pay much more attention to design elements of chatbot interface and familiarity of the brand that chatbot represent more (Praditomo et al., 2022). These incidental cues may either support or challenge their perception of the chatbot's reliability, accuracy, and trustworthiness, which in turn may

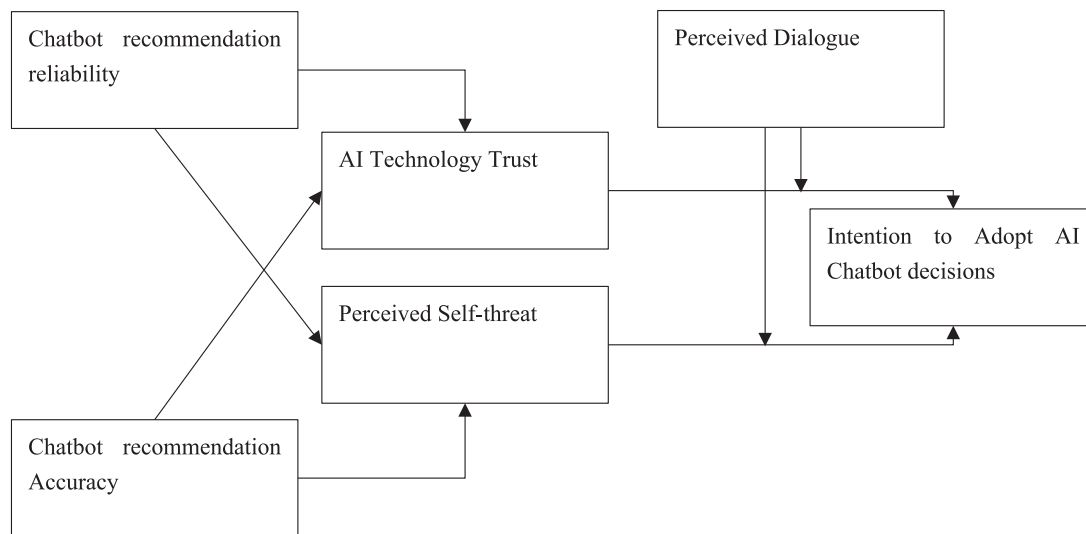


Fig. 1. Proposed research model.

realign their tendency to use the service. Similarly, individual with not enough cognitive ability might be also more prone to the effect of social norms or suggestions from the peers which also work as cue in their decision-making of whether to use human companion or the chatbot. Consequently, it is necessary to draw the ways helpful cues' play with cognitive restrictions for ease of AI chatbots adoption completeness analysis.

The central and peripheral mechanisms, thus, demonstrate distinct functionality dispelling existence of emotions and cognitive systems at play (C.-Y. Li, 2013). Notwithstanding the fact that individuals have either low or high EL and also the presence or absence of central cues, people tend to use these paths simultaneously. Nevertheless, there is evidence showing that with higher EL, the central pathway of information processing is more likely to be used, while with the lower EL, the peripheral pathway is more frequently chosen (Petty & Briñol, 2011). This support that both cognitive and emotional processing are available for the building of attitude changes.

2.2. ELM in AI chatbot recommendation context

Among the key AI chatbot recommendation characteristics of ELM, these types of AI bot recommendations show us how users give out their evaluations and reactions to recommendations. Central route processing, in other words, delves into factors of chatbot reliability, accuracy, and relevance to individual users' goals or objectives (Bitner & Obermiller, 1985). Higher EL persons will meticulously review the provided reasons and proof which will help them to make right choices as far as taking the recommendations of chatbots is concerned (Gong et al., 2021). Contrary to this, the peripheral route processing may entail a shallow exam through cues such as chatbot design appeal, brand familiarity, among other EL among users with low cognitive abilities (Petty & Cacioppo, 1984). The ELM allows us to fully grasp, from a psychological point of view, the basis of the users' reasoning and the way they will react to the recommendation of products (Weiler et al., 2022). The most striking feature of this process is the central ideas of recommendation reliability and accuracy. These will dictate both the attitudes and intentions of the users to use AI chatbots for product recommendations.

The main components of central route processing in the context of AI chatbot recommendations are based on the opinions and feedback of recommendation reliability and accuracy (Amatriain et al., 2009; Moradi & Ahmadian, 2015). These indicators lie at the core of those customers' attitudes and intentions just to accept AI robots as tools for suggestive product acquisition. The reliability of recommendations is

the perceived consistency and dependability of the advised information given by the chatbot (Goodman et al., 2023). Because, users tend to consider as authentic and trustworthy, the advice that is being given by AI chatbots that can do things like providing quick and accurate answers according to previous demonstrated abilities. It is important that recommendation reliability has quality factors such as machine learning algorithm correctness, accuracy in providing recommendations, and trust in the process of making recommendations (Wilkinson et al., 2021). Consumers care a lot about the reliability of chatbot recommendations when it comes to the AI chatbot because when it is reliable, they will have faith and this will make the likelihood of them adopting it as a trusted advisor with high decision processes. In the next place, it is important to talk about recommendation accuracy - this is the term that describes precision and relevance of the suggestions made by the AI chatbot (Amatriain et al., 2009). Well-designed recommendations are specifically targeted to each individual user's preferences, needs, and past interactions, thus providing a better alternative than the one-size-fits-all approach (Narducci, de Gemmis, Lops, and Semeraro, 2018). People appreciate the use of AI that gives them accurate and individualized suggestions that are customized to fit their personality, which in turn leads to their increased joy and engagement with recommendation systems. Chatbots which provide precise responses to customers build a commendable image among them and this creates more confidence in them to seek for solutions from the chatbot on future happenings. Consequently, in the case of the central route processing, the trustworthiness and correctness of chatbot recommendations produced by the AI serve as the key prerequisites for the users' recognition and adoption of AI-filled recommendation systems. Organizations and developers should consider these main cues as a top consideration while developing chatbots that are interesting and reliable in presenting quality and trustworthy product recommendations to consumers.

The objective of our study is to enrich the existing ELM framework by exploring the various components of such chatbot recommendations, namely, their reliability and accuracy, for the purpose of evaluating their impact on users' recognition and intention to adopt the new technology. We suggest that the users' trust in AI technology and its threat towards one's job, alongside the chatbot's perceived dialogue efficiency are factors essential in the formation of elaboration likelihood and decisions by users (Aghakhani et al., 2021; Chen & Barnes, 2007; Leong et al., 2019).

These variables formed into the compound of the ELM model (i.e. AI technology trust, perceived self-threat, reliability and accuracy of recommendations) are the focus in the study and they help us to have a more complete knowledge about the processes that regulate AI chatbot

recommendations acceptability by the consumers. We are confident that in the case of AI technology, users will trust it and their perception of self-threat will influence how likely they are to choose a situation where they are processing information through the central route as they give their judgment on the recommendations provided by chatbots. Furthermore, we postulate that the perceived dialogue of the chatbot will moderate the relationship between recommendation honesty and users' elaboration of thoughts.

A combination of both empirical data and analytical research is being used to test for these hypothesized relationships, ultimately providing empirical support for this claim and adding theoretical depth to this topic. Providing AI chatbot designing with the cognitive and emotional aspects of user decision-making offers a possibility to improve the effectiveness of AI chatbots while engaging users and building trust that ultimately contribute to the customer satisfaction and business results (Ashfaq et al., 2020).

2.3. Research model and hypothesis development

2.3.1. Chatbot recommendation reliability and accuracy relation with AI technology trust and perceived self-threat

As has been mentioned earlier, in the past few years AI-based chatbots have developed a very important role to play in the tasks of personalized recommendations in domains that concern e-commerce and customer service (Nichifor et al., 2021; Rakhra et al., 2021; Shafi et al., 2020). As users relying more and more on the chatbots to help to reach their decisions this reliability of the recommendations made by the AI becomes the main factor in determining the trust in AI technology (Kim et al., 2021). Here reliability entails the level of correctness and quality of the chatbot recommendations which are offered (Lechien et al., 2024). Research has proven that recommendation reliability is a major factor that impacts on users' perception of AI technologies, and trusted recommendation of AI systems may increase users' trust and confidence in the capabilities of chatbots (M. Li & Wang, 2023). Consequently, the interdependency between reliability of recommendation and trust in AI technology plays a crucial role in the process of improving user's acceptance and following of AI-driven recommendation systems (Bedué & Fritzsche, 2022).

Trusting in AI systems is a must to enable the application of AI technology in multiple areas and even expansion to other areas. Trust involves the achievement of confidence by users in AI systems with the regards of dependability, effectiveness and benevolence (Venkatesh et al., 2003). The capability of the chatbot's recommendations is very essential in developing confidence with the user as it infers his capacity to give precise and consistent suggestions which suit the consumer's choices and desires. One of the findings is that users are more likely to react positively to such AI systems that give dependable suggestions and thus create higher engagement (De Cicco et al., 2021). In this respect, evaluating the connection between recommendation reliability and AI technology trust logically leads to the identification of the factors influencing trust in AI driven recommendation systems. However AI technology, apart from the reliability of chatbot suggestions, users are now expanding their opinions to issues of the system's transparency, accountability, and ethical character (Chen & Barnes, 2007). Although recommendation reliability is defined as one of the key aspects in enhancing the trust, users nevertheless often trust AI technologies with respect to other factors namely, algorithm transparency, data accessibility and user feedback (Mostafa & Kasamani, 2022). In other words, determining the correlation between the reliability of recommendations and trust in AI technology brings up the duality of trust in AI-powered recommendation systems and leads to the implementation of strategies that increase the trustfulness level and acceptance by users.

Similarly, the accuracy of chatbot recommendations means that the suggestions offered to users must be enough to the point and be relevant to their needs or preferences (Narducci et al., 2018; Suhaili et al., 2021). In a nutshell, trust in AI technology refers to user's belief in the quality,

expertise, and neutrality of these systems (Venkatesh et al., 2003). Research indicates that accuracy in recommendation, precisely, was the factor that affected the most to the trust in AI technology perceived by the user (Goodman et al., 2023). Extensive literature has explored the role of dialogic communication in developing customer trust in chatbot services (Jiang et al., 2022; Tsai et al., 2021). Human qualities including responsiveness and ability to engage in conversations assists users gain confidence while interacting. This creates a favorable feeling and interaction with chatbots (Araujo et al., 2018). Research has also indicated that using naturally designed chatbot dialogs that replicate human-to-human engagements improves communication outcomes, fosters trust, and solidifies social interactions. Users also feel a higher level of perceptual congruency with virtual agents when they offer hint-prompting cues and engage users in conversational prompting (Rese & Tränkner, 2024). Lastly, customers should have the confidence that there are always chatbot agents ready to assist them, willing to converse and able to hold a conversation. This consent rate is directly connected with the chatbot acceptance rate. When customers recognize that AI chatbots can deliver prompt, suitable, and personalized recommendations to their liking and preferences, they will perceive the technology as accurate (Amatriain et al., 2009). Consequently, the logic behind the association between recommendation accuracy and AI technology trust is what determines the significant factors of the user's effort to trust and rely on AI-driven recommendation systems. Therefore in light of the discussion and scientific evidence we can hypothesize that:

H1. There is a positive relationship between chatbot recommendation reliability and AI technology trust.

H2. There is a positive relationship between chatbot recommendation accuracy and AI technology trust.

The perceived self-threat include users' fear of job losing from automation, no autonomy, or losing decision control owing to dependence on AI (Chen & Barnes, 2007). Perceived threat relating to chatbot can be defined as the extent to which people are able to evaluate the risk or danger that may be posed by chatbot (Ho et al., 2018). The scholars identified that the end-users are more likely to include chatbot suggestions when making decisions and accept AI more as opportunity than threat when they perceive the gains (Jiang et al., 2022; Pillai et al., 2024). Reliability of chatbot recommendation- how trustworthy, consistent and reliable its recommendations are- is important to create positive perception about self-threat in user's minds. One of the reasons users find chatbots recommendations trustful and reliable is their ability to show confidence and consistence (Baek & Kim, 2023; Campbell & Sedikides, 1999). In this manner, users are not likely to experience worries and hesitation about decision making processes. Thus, looking into the impact of recommendation genuineness on self-threat perception is inevitable for us to understand the depth of chatbot suggestions and the way they influence a user's perception of job displacement and autonomy. Concern of recommendation accuracy includes the parameters like precision and relevance of chatbot recommendation to users' needs and wishes (Goodman et al., 2023). According to Wang et al. (2023), the latest research on the role and impact of recommendation accuracy in customers' belief and their trust is a vital factor. Users with the notion that chatbot suggestions seem precise and aligned with their needs, the insecurity and apprehensions in his mind about taking AI technology decisions is very often to occur. Therefore, researching the relationship between recommendation accuracy and perceived self-threat is of extreme importance for understanding the effects of accurate recommendations of chatbot users on their work replacement fears and freedom of choice. Moreover, AI development and integration in various sectors will necessitate determining how the reliability and accuracy of chatbot recommendations affects user perceptions and actions, thus information gathered may lead to trust, acceptance, and engagement. Through overcoming self-threat of the users and upgrading the mode of the chatbot into more reliable and accurate recommendation

system, AI technology may reduce some concerns of the users and even gladden them further (Brown & Halpern, 2021). Therefore, we propose that:

H3. There is a negative relationship between chatbot recommendation reliability and perceived self-threat such that chatbot recommendation reliability reduces the perceived self-threat.

H4. There is a negative relationship between chatbot recommendation accuracy and perceived self-threat such that chatbot recommendation accuracy reduces perceived self-threat.

2.4. AI technology trust and perceived self-threat relation with intention to adopt AI chatbot decisions

The influence of AI technology trust to AI chatbot adoption decision has an effect that is meaningful and multidimensional. Trust in AI technology is a measure of the users' degree of conviction of the accuracy, effectiveness, and ethical behavior of AI systems (Choung et al., 2023). People are more likely to make decisions upon the word of an AI chatbot when they believe that the technology is trustworthy enough to generate the predicted and useful results (Siau & Wang, 2018). A research work conducted by Ryan (2020) has indicated that trust in AI could contribute to the acceptance of AI-driven services, similar to chatbots among users by increasing their intention of adopting those services. This is so because trust causes to avoid risks linked to technology adoption by reducing privacy, security, and accuracy doubts. Subsequently, the number of people who are keen to interact with AI chatbots increases thus, higher usage of AI chatbots results in high usage rates of AI solutions in various applications (Adam et al., 2021).

On the flip side, people take the notion of self-threat to be a daunting obstacle to the ability to follow AI chatbot conclusions (Hasan et al., 2023; Song et al., 2022). Individuals fear that AI would pose a threat to their self-worth by superimposing its competence and efficacy, consequently making them feel inadequate, lack of autonomy and endangering their job security (Yu, 2021). Since AI ages at a distressing speed and is able to carry out tasks that have always been done by human beings, people may start to perceive their jobs being affected by robots. Likewise, workers in the customer service fields can be expected to erect a barrier to AI-based chatbots and solutions as they are afraid to be ousted from their jobs, thus the resistance and reluctance may arise. In the research of Bigman et al. (2019), the authors suggest that actually perceived self-threat (real or imagined) has a negative impact on how people feel about AI systems and the likelihood whether or not to trust AI technologies. Thus, rectifying the issues linked directly to self-threat is essential in the execution of AI chatbots' decisions and the development of pro-attitudes to AI progress.

Empirical data demonstrate that a positive relationship exists between AI technology trust and AI chatbot decision adoption intention (Choudhury & Shamszare, 2023; Mostafa & Kasamani, 2022). Trust that serves as one the most critical predictors of AI technology adoption and utilization, is an important factor for users to assess whether AI is reliable, credible, and useful. This shows that trust and adoption significantly influence the intention to adopt and use new technologies, thus emphasizing the role of trust in the decision to adopt technology (Gefen et al., 2003). Consequently, the trustworthiness of the technology can be increased by using transparent and accountable AI chatbots that are properly communicated with and, hence, will not impact the users in a negative way. Contrary to the higher trust in the AI technology, the perceived self-threat exhibits a negative correlation regarding the intention to endorse the AI chatbot decisions (Chatterjee et al., 2002; Liu & Tao, 2022). Individuals believing that AI is a threat to their self-perception and professional competence are more likely to reject AI implementation and will remain loyal to the conventional approach, in which people are the first in line. Along these lines, Zhang et al. (2014) highlighted the need to address the perceived self-threats involved in issues about job displacement, skill obsolescence, and job decision-

making processes. Stressing the fact that AI can work alongside humans instead of replacing them, organizations will be able to respond to the self-conscious threat and create a friendly climate concerning the interplay between AI and humans. Therefore, trusting the AI chatbot takes the lead but there is the need to address the threat in order to overcome the psychological barriers to the adoption, hence, developing a supporting environment for the AI advancement. Therefore we hypothesize as:

H5. There is positive relationship between AI technology trust and intention to adopt AI chatbot decisions.

H6. There is a negative relationship between perceived self-threat and intention to adopt AI chatbot decisions.

2.5. Moderating role of perceived dialogue

The perceived dialogue as a key facet or element is considered a powerful moderator shaping the connection between the trust in AI technology and the willingness to adopt AI chatbot decisions (Tsai et al., 2021). Perceived dialogue is a human factor that relies on the comprehension users form about interactions, naturalness, and responsiveness between them and the AI chatbots (Gerald, 2009). When users think that chatbot conversations are interesting, accurate, and customizable to their specific requests, they are more likely to trust the AI system that is behind that communication. According to Bickmore and Picard (2005), the factor that can help build trust and positive impressions from the users is whether they feel as if they are talking to someone or an artificial intelligence. The employment of reliable dialogue strategy likely increases the user's perception of AI competence and reliability that in turn results in augmented their trust in AI-based alternatives (Chang & Chuang, 2011). Further, interacting communication enables the sharing of information, problem-solving, and decision-making processes and as a result of this it adds to the concept of usefulness and value of AI chatbots in supporting users with doing different things and answering some questions (Venkatesh et al., 2003). Subsequently, humanizing the interaction process via textual analysis, context empathy, and natural language processing can be the key which will help organizations to build a bridge between AI technology trust and the decision to adopt AI chatbot decisions which will lead to increased user acceptance and engagement (Pavlou, 2002).

Moreover, the moderating role of perceived dialogue which lies between the perception of self-threat and the intention to accept the AI chatbot's decisions is also highlighted. Individuals who interpret chatbot interactions as supportive, respectful, and user-centered, are less likely to feel diminished by the AI technology or threatened by job displacement or skill obsolescence, but are likely to accept the AI chatbots, considering the associated benefits. Campbell and Sedikides (1999), shows that an emphatic and personalized conversation can lessen the perception of self-threat by stressing on the cooperation and collaboration between the AI and human beings instead of making the AI to replace human being without a role. Through nurturing the sense of coexistence and collectiveness between machines and humans the image of dialogue within the AI-chatbots alleviates the fear that people might give up on important decisions or fail to obtain the social recognition and respect in the present society (Zirar et al., 2023). Furthermore, opening up dialogue creates an opportunity by which users can articulate their desires, give their opinions, and have a say on how AI chatbots behave. This way, the users feel more comfortable and in charge of the conversations they have with AI technology. Hence, by giving more attention to user centered design practices and implementing empathetic discourse strategies, the organizations can eliminate the possible self-threatening perceptions and elicit the acceptance of AI chatbots decisions, with this, it can be said that there is harmonious integration of human and machine in decision making process.

Hence, it can be inferred from the above discussion that:

H7. Perceived dialogue positively moderates the relationship between AI technology trust and intention to adopt AI chatbot decisions, such that this relationship is stronger in presence of perceived dialogue.

H8. Perceived dialogue negatively moderates the relationship between perceived self-threat and intention to adopt AI chatbot decisions, such that the relationship is weaker in presence of perceived dialogue.

3. Methods

In light of our broad research premise, the survey method was therefore considered the most appropriate and logical for the following reasons. Firstly, e-commerce is the most common sector that utilizes AI chatbot where product suggestions are the major components of chatbots in the buying processes online. As a result, customers predominantly interact with AI chatbots to ask for product recommendations. Consequently, the survey approach is a suitable means to identify eligible respondents. Secondly, the parameters in the study framework refers to the persuasion process of consumers psychologically and hence there can be an examination of the types of instruments that capture the consumers psychologically. The model of this study was tested using an online survey method. Following previous research (Ali et al., 2019; Cao et al., 2021), we used validated measures by a comprehensive literature review. We collected data from Chinese consumers who frequently use e-commerce websites for shopping. This context is appropriate to test our model because AI chatbots are being widely employed in online shopping context and customers commonly interact and receive information regarding products from these AI chatbots (Han et al., 2022). We used an approach to ensure that respondents have experience of using AI chatbot recommendations. Initially, we added a statement in the questionnaire suggesting this survey is designed for individuals who have used AI chatbots and received recommendations from it. Therefore, please do not respond this survey if you have not used it or if you have used but did not receive any recommendations. Before administering the large-scale survey, five academic scholars specializing in e-commerce assessed the content validity of the survey instrument. They recommended several changes to the questions, including adjustments to phrasing, content, order, structure, and presentation.

We shared link to online groups using snowball sampling approach. After a month of survey, we received 531 completed responses. In an initial screening we deleted 29 respondents who reported same answer to all items. Among 411 useful responses, there were 278 (67.6 %) females and 133 (32.3 %) males. Age of the most respondents was between 18 and 35 (83.2 %). Most respondents hold (53.3 %) master's degree.

Self-reported data can potentially get affected by common method bias (Ali et al., 2020). Therefore, we performed a Harman's single-factor analysis to test whether the data is free from the issue of common method bias. Results demonstrated that the first factor accounts for 30.23 % of the variance, and 7 factors account generate value >1. These results suggest that common method bias did not affect findings of our study (Podsakoff et al., 2003). In addition, prior research has suggested that moderation results are not significant if data is affected from common method bias. Our significant moderation results provide further support that common method bias is not a significant issue in our data. Additionally, we conducted a multicollinearity assessment using the variance inflation factor. The results showed that all variance inflation factors were below 10 (Kock, 2015), indicating that the data did not present any multicollinearity issues.

3.1. Measures

All measures of the study were adopted from previous studies presented in Table 1. The survey items of this study were slightly changed to appropriate fit in the context of this study. A 7-point Likert scale (1 = strongly disagree to 7 = strongly agree) was employed to record the responses of the participants. The questionnaire was initially prepared in English and then later

Table 1
Measures.

Variable	Item	Measure	Source
Chatbot recommendation reliability	RRE1	I believe that the products that the AI chatbot recommended are reliable	Devaraj et al. (2002)
	RR2	I believe the AI chatbot recommends me the products I asked for from him	
	RR3	I think that the AI chatbot I interacted with performs the service of recommendation right	
Chatbot recommendation accuracy	RAC1	In general, most items recommended by the AI chatbot match my preferences very well	Zhang et al. (2011)
	RAC2	In general, most items recommended by the AI chatbot fit my tastes very well	
	RAC3	In general, most items recommended by the AI chatbot are interesting to me	
AI Technology Trust	AIT1	I trust chatbots to be reliable	Chandra et al. (2010)
	AIT2	I trust chatbots to be secure	
	AIT3	I believe chatbots are trustworthy	
	AIT4	I trust chatbots	
	AIT5	Even if the chatbots are not monitored, I would trust them to do the job correctly	
Perceived self-Threat	PT1	I do not feel totally safe providing personal private information over chatbots	Wong et al. (2024)
	PT2	I am worried about using chatbots because other people may be able to access my account	
	PT3	I do not feel secure sending sensitive information across chatbots	
	AID1	I am willing to use this AI chatbot as an aid to help with my decision about which product to buy	
	AID2	I am willing to let this AI chatbot assist me in deciding which product to buy	
Intention to Adopt AI Chatbot decisions	AID3	I am willing to use this AI chatbot as a tool that suggests to me a number of products from which I can choose	Komiak and Benbasat (2006)
	PD1	I felt like I was engaged in an active dialogue with the virtual agent	
	PD2	My interactions with the virtual agent felt like a back-and-forth conversation	
Perceive dialogue	PD3	The virtual agent responded quickly to my inputs and requests	Sundar et al. (2016)
	PD4	The virtual agent was efficient in responding to my activities	

reviewed by a professor of English department in order to ensure that the survey items were devoid of any linguistic inaccuracies. In an attempt to mirror this survey for the Chinese response, three Chinese people who are also experts in English language were hired to interpret the questionnaire from English to Chinese language (Van de Vijver et al., 1997). An experienced translator who had no clue about the original English version of the survey items then translated the Chinese version back into English language. The analysis of the semantic differences between the two survey items did not show any evidence of substantial differences. To ensure that the instrument was more reliable, two professionals who have a working experience in AI technology scrutinized

the items in the survey. Each respondent's feedback was analyzed and incorporated into modifying the particular measurement items as well as the entire survey layout to enhance coherence, conciseness, and efficiency.

3.2. Analysis and results

Analysis was performed in two steps following prior research (Bao et al., 2020; Cao & Ali, 2018; Islam et al., 2020; Tariq et al., 2023). We first analyzed measurement model to assess the reliability and validity of the dataset. In second step, we analyzed the structural model to test the hypothesized relationships of the study. In doing so, we used AMOS V 24.0 and PROCESS macro to test direct, mediation, and moderation hypotheses.

3.3. Measurement model analysis

We conducted an exploratory factor analysis and reported results in Table 2. Results reveal that factor loadings ranging between 0.55 and 0.95 are within acceptable threshold. These values suggest a reasonable discriminant validity. Table 2 further shows that values of Cronbach's α ranging between 0.76 and 0.90 and the composite reliability (CR) between 0.76 and 0.91. These values evidence of reliability of measurement model (Nunnally & Bernstein, 1994). Table 2 shows that values of average variance extracted (AVE) are all above 0.50, suggesting convergent validity of the measurement model (Davis, 1989).

We then calculated the square root of AVEs and the items' correlations. Results in Table 3 show that correlations are all low than the corresponding square roots the AVE. These results provide further evidence of discriminant validity of the measurement model. Additionally, the results of confirmatory factors analysis generate a reasonable value of fit indices (chi-squared/degrees of freedom (χ^2/df) = 1.57; the root mean square error of approximation (RMSEA) = 0.04; the comparative fit index (CFI) = 0.97; the Tucker–Lewis index (TLI) = 0.97. When $\chi^2/df \leq 3$, RMSEA ≤ 0.08 , CFI ≥ 0.90 , TLI > 0.90) (Bagozzi et al., 1991). Collectively, the analyses indicate that our measurement has acceptable level of validity and reliability statistics.

3.4. Structural model analysis

Results of structural model analysis are reported in Table 5. Hypothesis 1 predicted a positive relationship between chatbot recommendation reliability and AI technology trust. Accordingly, results demonstrate that there is a positive relationship between chatbot recommendation reliability and AI technology trust ($\beta = 0.38$, SE = 0.04, $p \leq 0.001$), supporting for Hypothesis 1. Hypothesis 2 predicted that there is a positive relationship between chatbot recommendation accuracy and AI technology trust. Results in Table 4 show a positive effect of chatbot recommendation accuracy on AI technology trust ($\beta = 0.21$, SE = 0.05, $p \leq 0.001$). Thus, results support Hypothesis 2.

Hypothesis 3 predicted that there is a negative relationship between

chatbot recommendation reliability and perceived self-threat such that chatbot recommendation reliability reduces the perceived self-threat. Results provide support for this relationship and suggest that there is a negative relationship between chatbot recommendation reliability and perceived self-threat ($\beta = -0.18$, SE = 0.06, $p \leq 0.001$). Similarly, Hypothesis 4 suggested that there is a negative relationship between chatbot recommendation accuracy and perceived self-threat such that chatbot recommendation accuracy reduces perceived self-threat. Results in Table 4 demonstrate negative effect of chatbot recommendation accuracy on perceived self-threat ($\beta = -0.40$, SE = 0.05, $p \leq 0.01$). Thus, Hypothesis 4 is supported.

Hypothesis 5 predicted a positive relationship between AI technology trust and intention to adopt AI chatbot decisions. Accordingly, results demonstrate that there is positive relationship between AI technology trust and intention to adopt AI chatbot decisions ($\beta = 0.37$, SE = 0.06, $p \leq 0.001$), supporting for Hypothesis 5. Hypothesis 6 predicted a negative relationship between Perceived self-threat and intention to adopt AI chatbot decisions. Results in Table 5 show a negative relationship between perceived self-threat and intention to adopt AI chatbot decisions ($\beta = -0.16$, SE = 0.05, $p \leq 0.001$). Thus, results support Hypothesis 6.

Prior to analyze moderation relationship, we mean centered the independent and moderating variables. Moderation Hypothesis 7 argued that perceived dialogue positively moderates the relationship between AI technology trust and intention to adopt AI chatbot decisions, such that this relationship is stronger in presence of perceived dialogue. Results in Table 5 demonstrate the interaction effect of perceived dialogue and AI technology trust on intention to adopt AI chatbot decisions is positive ($\beta = 0.12$, SE = 0.05, $p \leq 0.01$). We employed simple slope test to further assess the moderating effect. Fig. 2 shows that interaction effect. Results of simple slope test reveal that relationship between AI technology trust and intention to adopt AI chatbot decisions is stronger at high level of perceived dialogue ($\beta = 0.42$, SE = 0.06, $p \leq 0.001$) than when it is low ($\beta = 0.24$, SE = 0.06, $p \leq 0.001$). These results provide support for Hypothesis 7. Hypothesis 8 argued that Perceived dialogue negatively moderates the relationship between perceived self-threat and intention to adopt AI chatbot decisions, such that the relationship is weaker in presence of perceived dialogue. Results in Table 5 demonstrate the interaction effect of perceived dialogue and perceived self-threat on intention to adopt AI chatbot decisions is negative ($\beta = 0.11$, SE = 0.05, $p \leq 0.05$). We employed simple slope test to further assess the moderating effect. Figs. 2, and 3 shows that interaction effect. Results of simple slope test reveal that relationship between perceived self-threat and intention to adopt AI chatbot decisions is weaker at high level of perceived dialogue ($\beta = -0.18$, SE = 0.06, $p \leq 0.01$) than when it is low ($\beta = -0.35$, SE = 0.06, $p \leq 0.001$). These results provide support for Hypothesis 8.

4. Discussion

The research findings brought out the complex understanding of the interplay between various factors that determine the users' intention to follow the AI chatbot advice. Firstly, the results found that chatbot recommendation reliability, accuracy, and AI technology trust all have strong positive relationships. Customers that relate AI recommendations to being trustworthy and precise are more likely to have faith in the technology (Goodman et al., 2023; Moradi & Ahmadian, 2015). This confidence comes from users' trust that AI system services are both competent and credible, showing how important recommendation quality is in improving technology trust. Furthermore, the study demonstrates that individual's perceive chatbot recommendation reliability and accuracy inversely with the perceived self-threat. The user reaches a point where she or he experiences a reduction in perceived self-threat along with the increase in recommendation reliability and accuracy. Accurate and trustworthy recommendation reduces the concerns of individuals about mistake and bias, which result in the lessening of self-

Table 2
Results of confirmatory factors analysis.

Variable	CR	AVE	Cronbach's alpha	loadings
Chatbot recommendation reliability	0.76	0.52	0.76	0.69–0.74
Chatbot recommendation Accuracy	0.79	0.56	0.78	0.67–0.87
AI Technology Trust	0.83	0.50	0.83	0.55–0.85
Perceived Self-threat	0.80	0.57	0.80	0.73–0.80
Intention to Adopt AI Chatbot decisions	0.80	0.57	0.80	0.69–0.85
Perceived Dialogue	0.91	0.71	0.90	0.63–0.95

Note: CR = composite reliability, AVE = average variance extracted.

Table 3

Means, standard deviations, and correlations.

Variable	Mean	Std. Deviation	1	2	3	4	5	6	7	8	9
Chatbot recommendation reliability	4.84	0.71	0.72								
Chatbot recommendation Accuracy	5.88	0.63	0.01	0.75							
AI Technology Trust	5.28	0.65	0.42**	0.21**	0.71						
Perceived Self-threat	3.60	0.84	−0.34**	−0.14**	−0.38**	0.76					
Intention to Adopt AI Chatbot decisions	4.93	0.84	0.23**	0.17**	0.34**	−0.26**	0.76				
Perceived Dialogue	2.81	1.02	0.04	0.05	0.04	0.01	0.14**	0.84			
Gender	1.32	0.47	0.07	−0.01	0.00	−0.07	0.01	0.07	–		
Age	1.70	0.79	0.09	0.00	0.09	0.00	0.00	0.01	−0.17**	–	
Education	3.35	0.60	0.00	−0.01	0.03	−0.07	−0.01	−0.03	−0.02	0.117*	–

Note:

* $p \leq 0.05$,** $p \leq 0.01$,

threat perception that will promote the development of a more positive view of AI chatbots. Moreover, the results indicate positive associations between AI technology trust and intentions to adopt AI chatbot decisions, as well as negative associations between perceived self-threat and intentions to accept AI chatbot decisions. The trust levels in AI technologies are directly proportional to the intention of the users in embracing AI chatbot outputs, showing the criticality of trust in shaping users' adoption attitude. Contrarily, the higher degree of perceived self-threat, the less intention to adopt AI chatbots has been reported, highlighting the importance to dispel users worries in the process of a transition to AI systems.

Furthermore, the study discusses the moderating effect of perceived dialogue. The perceived dialogue moderates the relationship between AI technology trust and intent of people to accept AI chatbot decisions in a positive manner. In other words, the presence of perceived dialogue supporting the technology boosts up the positive association between trust in AI technology and intention to accept the decision. On the one hand, perceived dialogue is a moderator that negatively functions the association between perceived self-threat and intention to adopt AI chatbot decisions, and it implies that the existence of dialogue weaken the negative impact of perceived self-threat on adoption intention. In general, it indicates that the recommendation is of paramount importance for the AI technology trust, self-threat perception, dialogue, and ultimately decision adoption intentions. These findings have serious consequences for creating and implementing AI chatbot systems that ensure credibility, precision, and user trust as well as addressing users' concerns and engagement enhancement through dialogue process.

4.1. Theoretical implications

Theoretic results of this study are plausible not only in the area of technology adoption and human-computer interaction, but they could be used by researchers, professionals, or policymakers in different fields of their interests. To begin with, the study is linked to the ELM framework through the elucidation of its applicability in the domain of AI chatbot recommendation systems. Through the use of the ELM principles, the study illuminates the cognitive processes that underlie the way users decide to go for AI chatbots adoption by implying that emphasis should be put on central cues such as recommendation reliability and accuracy in the way users shape their attitudes and behaviors. Our study broadens the scope of theoretical studies on persuasion and attitude change under the umbrella of ELM to include elaboration processes, which are especially crucial in the context of AI technology adoption. The subsequent research might expand on the importance of peripheral cues and heuristic processing models in how human beings interact with Artificial Intelligence chatbots. This would advance our knowledge on the persuasion mechanisms in technology-mediated environments.

In addition, the study delivers on trust theory in that it focuses on the role of AI technology trust in impacting people's willingness to adopt AI chatbot decisions. The results draw our attention to the school of

thought that suggests that trust is the heavyweight among all the drivers of technological adaptation, in particular, those related to the application of artificial intelligence. The study aims at determining the connection between the accuracy, reliability, and innovativeness of chatbot recommendations to the level of the trust that users develop in AI technology by enhancing the understanding of the crucial role of perceived reliability and competence in building such trust. This shows trust theory through empirical evidence of the truth that trust is the decisive factor for technology adoption. Subsequently, it can benefit for the investigations of effective strategies to build trust for AI chatbot systems.

Furthermore, the research extends the context to threat appraisal theory by exploring how cognizance of personal threat influences the propensity of the user to adhere to the AI chatbot verdicts. The study does this by looking at how recommendation reliability and accuracy affect users' beliefs about self-threat, which then extends threat appraisal theory to the domain of technology adoption, thereby uncovering the psychological mechanisms that drive users' responses to risks and uncertainties pertaining to AI chatbots. Such a theoretical insight helps us to comprehend how users assess the advantages and disadvantages of integrating the AI driven systems to the extent and emphasizes the necessity of taking measures to nip perceived self-threat and others negative perception in the bud in order to encourage technology acceptance and adoption.

Lastly, the research also adds to the existing dialogue systems literature by studying how the perceived dialogue moderates the interplay between AI technology trust, perceived self-threat, and the likelihood to adopt AI chatbot recommendations. By delving into how the incorporation of dialogue into human exchanges with AI chatbots can affect trust and threat perceptions of the users, the study adds to our knowledge of the role that dialogue plays in shaping user perceptions and behaviors in human-computer interaction. The conceptual understanding here has the potential for the construction and technological deployment of AI chatbot systems that utilizes dialogue mechanism to aid in driving user engagement, trust and intention to adopt the systems, which will in turn contribute to the improvement of dialogue system field of research and practice.

4.2. Practical implications

The implications of this research for the organizations, developers, and other AI system stakeholders cannot be underestimated as these professionals help in the design, development, and deployment of AI chatbots for different use cases. Thus, reliability and accuracy of recommendation system is the key factor which has to be addressed when designing the AI chatbot system. Developers are better advised on deploying exceptionally strong AI algorithms and establishing intuitive databases for the chatbot recommendations to be highly reliable and precise. This includes many approaches such as utilization of machine learning algorithms, natural language processing methods, and user

Table 4
Hypothesis testing results.

Hypothesis	Input	Outcome	Estimate	S.E.	Result
H1	Age	AI technology trust.	0.04	0.04	Supported
	Gender	AI technology trust.	−0.02	0.06	
	Education	AI technology trust.	−0.01	0.03	
	chatbot recommendation reliability	AI technology trust.	0.38***	0.04	
H2	chatbot recommendation accuracy	AI technology trust.	0.21***	0.05	Supported
H3	Age	Perceived self-threat	0.04	0.05	Supported
	Gender	Perceived self-threat	−0.09	0.08	
	Education	Perceived self-threat	0.07	0.04	
	chatbot recommendation reliability	Perceived self-threat	−0.18***	0.06	
H4	chatbot recommendation accuracy	Perceived self-threat	−0.40**	0.05	Supported
H5	Age	Intention to adopt AI chatbot decisions	−0.02	0.08	Supported
	Gender	Intention to adopt AI chatbot decisions	−0.03	0.05	
	Education	Intention to adopt AI chatbot decisions	0.03	0.04	
	AI technology trust	Intention to adopt AI chatbot decisions	0.37***	0.06	
H6	Perceived self-threat	Intention to adopt AI chatbot decisions	−0.16***	0.05	Supported
H7	Perceived Dialogue	Intention to adopt AI chatbot decisions	0.13**	0.05	Supported
	Interaction 1	Intention to adopt AI chatbot decisions	0.12**	0.05	
	Perceived Dialogue	Intention to adopt AI chatbot decisions	0.13**	0.05	
	Interaction 2	Intention to adopt AI chatbot decisions	0.11*	0.05	

Note: Interaction 1 = Perceived Dialogue x AI technology trust; Interaction 2 = Perceived Dialogue x Perceived self-threat;

* $p \leq 0.05$,
 ** $p \leq 0.01$,
 *** $p \leq 0.001$.

feedback for the purpose of keeping continually improving the recommendation quality. Such as by improving recommendation's reliability and accuracy, designers can increase user's trust in AI chatbots, and consequently, the utilization of the target users will increase.

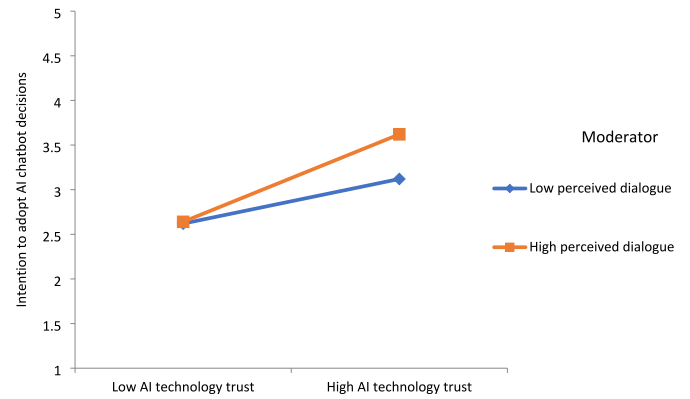


Fig. 2. Interaction effect of AI technology trust and perceived dialogue on intention to adopt AI chatbot decisions.

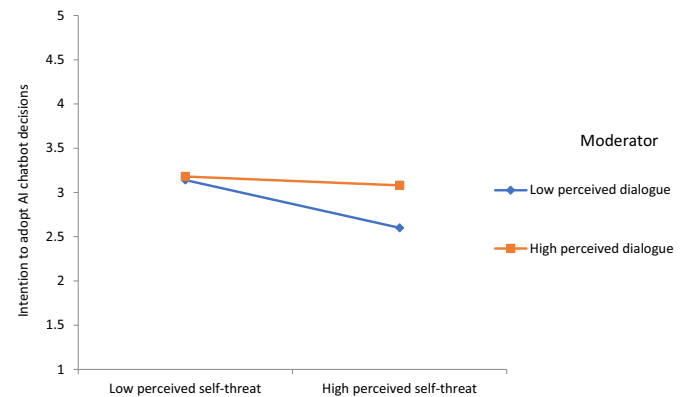


Fig. 3. Interaction effect of perceived personal-threat and perceived dialogue on intention to adopt AI chatbot decisions.

The second point, the necessity of clear communication strategies in AI chatbot interactions, is also emphasized in the study. It is necessary that AI chats provide users with clear, understandable explanations of how their decisions were made and the system's own boundaries. Transparency of the model through which recommendations are generated and the level of confidence associated with them can provide users with more confidence in bot capabilities. Practitioners will offer this through user-friendly interfaces, explanatory messages, and interactive features that allow users to check the reasoning behind the chatbot's recommendations and make comments on them. Transparent communication in that area can help creating a supporting environment for end-users regarding the threat to themselves and in turn encourages them to adopt AI chatbots.

Subsequently, the study emphasizes that user education and training programs are crucial for the acceptance of AI chatbots. Developers need to create such materials as manuals and interactive courses, which will be useful for teaching the users how to operate the AI chatbots, their advantages, and the limitations. This can be accomplished through tutorials, FAQs, and user guides, which helps people understand how to converse with the chatbot, how to interpret the recommendations and how to submit their feedback. The developers in this way can overcome user apprehensions, increase user confidence, and boost usage intention by empowering the users with knowledge and skills to effectively interact with AI chatbots. Besides, it is beneficial to develop training and support initiatives that are in line with feature and update introductions as an extension for the users to feel the motivation to engage more and adopt these innovations.

Finally, research has shown that individual user experiences are among the most important elements in the design of AI chatbots.

Developers need to consider personalization features that are adjusted to the individual preferences, requirements, and contexts of the users. This is achieved by tailoring the recommendations and interactions. It might be, of course, that the action involves such mechanism for personalization, data analysis, and contextual information that are used for adjustment of dialogue flows and recommendations in real time. Through personalized delivery, developers and organizations are able to improve customer satisfaction, engagement and user trust in the chatbots by AI thereby resulting in their having increased adoption and retention. Characteristics of personalization may involve personalized recommendations, adaptive dialogue prompts and proactive assistance that has taken into account user's behavior and preferences.

4.3. Limitations

This research has the following limitations which call for further explorations in the future. Primarily, this study takes a glance at a not yet streamlined area of AI chatbot recommendation processes. While future studies can confirm and expand these findings, it is worth noting the customization of AI chatbot platforms and individualization of its design in a wide range of contexts. For instance, multiple types of chatbots may involve different recommendation strategies that would make it mandatory to carry out thorough study to draw out meaning from multitudes of aspects. Secondly, the study neglects an assessment of the role of product type in the implementation of AI chatbot recommendation success. Through the investigations, different products or categories might have working recommendation approaches, which can further be implemented to get in-depth knowledge of the best and suitable recommendation content or delivery methods. Thirdly, the research only covers the pre-purchase communication step in the online shopping case study. For further studies, they may move to consideration of those scenarios where AI chatbots behave like the ones that specifically send recommendations outside of conversational contexts. It is necessary to know what recommendation characteristics are of great importance to customer adoption when such settings are considered. This is necessary as it could provide useful insight into the improvement of AI chatbot effectiveness. Finally, the study concerns mainly AI chatbots' position in pre-sales engagement; but many opportunities to discuss their application in post-sales service delivery could be considered in the future as well. A test of customer feedback on AI-based post sale services will help to address such issues as the opportunity to identify more areas for applying artificial intelligence in online retail to increase the overall level of customer satisfaction. Therefore, future researches should include those shortcomings to broaden the scope of Chatbots' capacities and its implications throughout the journey online shoppers make while traversing the different stages and settings of the experience.

5. Conclusion

This research delivers key insights into how AI chatbot recommendations affect online shopping decisions, using the elaboration likelihood model (ELM) to identify the most important behavioral drivers of user adoption intention. This study highlights the role of reliability, accuracy, trust in AI technology, and perceived self-threat as paramount factors in intention to adopt chatbot decision. The quality of recommendations and open and transparent communication should be among the top priorities, together with personalization and ethical aspects. Although the findings are fruitful in terms of bringing to table actionable suggestions to individuals, the study also pinpoints possible avenues for future research like studying the impacts of chatbots on different sectors, communication types, product categories and post-sales services. By covering these lacunae, experts will find out more about AI chatbot communication processes and how they influence the online consumer's behavior. Consequently, this research would add to the ever-changing narrative on human-computer interface, technology acceptance, and e-

commerce with real practical effects for stakeholders who desire to use AI chatbots to optimize user engagement and satisfaction in online shopping settings.

Ethics statement

This study was conducted in accordance with the ethical principles and guidance from ethical committee. All study protocols were approved by ethics committee of College of Liberal Arts and Science, University of Illinois Urbana-Champaign. The following is ethical number of study. ULUC-1401. Further, an informed consent was obtained from the individuals who have participated in this study.

CRediT authorship contribution statement

Angelina Lilac Chen: Project administration, Software. **Xinyang Piao:** Investigation, Formal analysis. **Manning Yu:** Project administration, Methodology. **Yakang Zhang:** Supervision, Software. **Lihao Zhang:** Writing – review & editing, Writing – original draft, Visualization.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

Data availability

The data that has been used is confidential.

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